

Pejman MERN Wonderland

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Background of the Project

This project is a **full-featured gaming and advertising web application** that combines entertainment, competition, and promotional opportunities in a single platform.

Users can play a variety of engaging games, earn stars, compete on leaderboards, promote themselves or their businesses, and in certain situations even generate income through advertising opportunities.

The application supports two types of users:

- **Players (Regular Users)**
- **Administrators (Admins)**



Player Features (Regular Users)

Account Management

- Register through a dedicated signup form with comprehensive validation.
- Automatically receive a personal profile page upon registration.
- Edit profile information through a validated profile-editing form.
- Customize the application's appearance by selecting different themes.

Gaming Experience

- Log in and enjoy **21 interactive games**, with **18 games offering multiple difficulty levels**.
- Earn stars by winning games and completing challenges.
- Compete with other players and climb the leaderboard.



Advertising & Visibility

- View advertisements displayed in a carousel format.
- View the **Top 10 Players** leaderboard, including player messages and star counts.
- Display a personal advertising message to all users by earning enough stars to reach the Top 10.

Community Features

- Submit ratings and reviews for individual games.
- Browse ratings and reviews from other users through paginated pages.
- Edit or delete personal ratings and reviews.
 - Editing includes form validation.
 - Deletion requires confirmation.

Security & User Experience

- Secure login and logout functionality.
- Logout action includes a confirmation prompt.



Administrator Features

Administrators have access to all player features, plus additional management capabilities.

Advertisement Management

- Create new advertisements through a validated form.
- Edit or delete advertisements they own.
 - Editing includes validation.
 - Deletion requires confirmation.

User Management

- View all registered users in a paginated table.
- Delete registered user accounts (with confirmation to prevent accidental removal).
- Perform **live user searches** based on username.
- Sort users by:
 - Username
 - Number of stars earned



Review Moderation

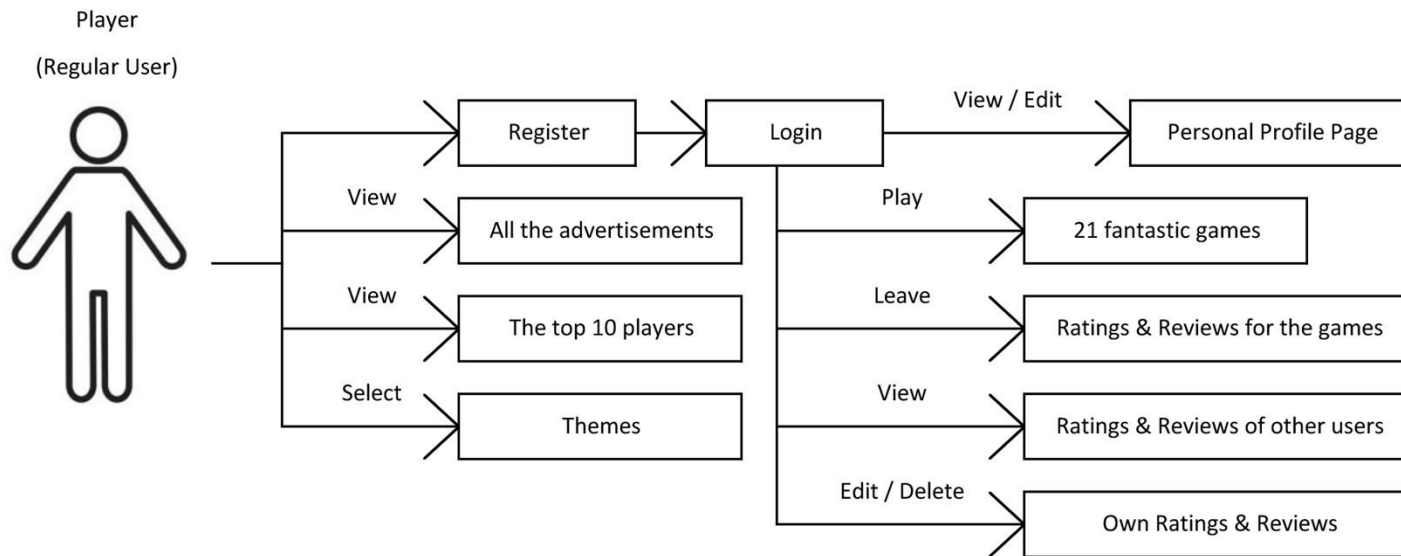
- Edit or delete **any** rating or review on the platform, not just their own.
 - Editing includes validation.
 - Deletion requires confirmation.

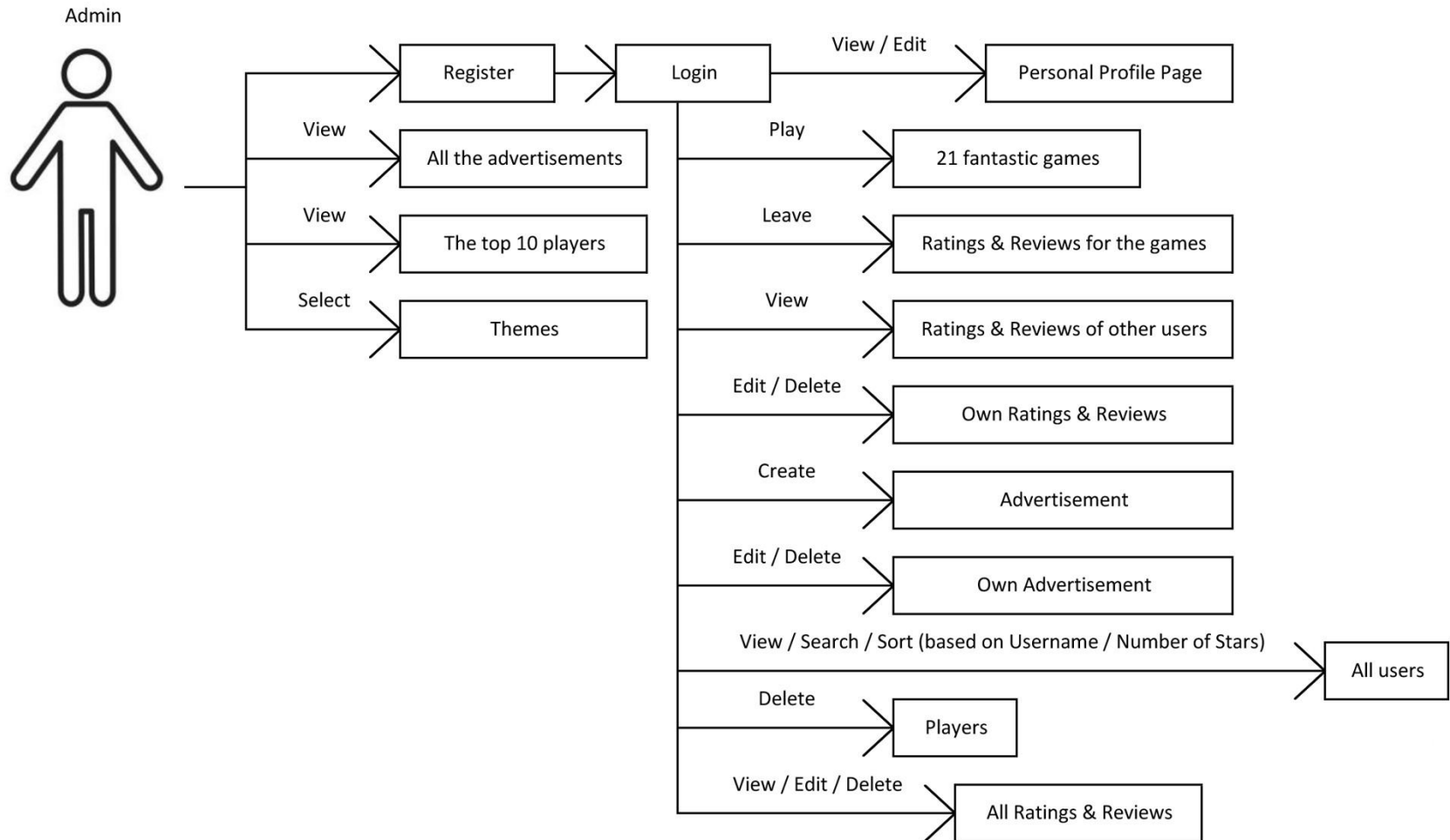
Key Highlights

- ✓ 21 interactive games (18 with multiple difficulty levels)
- ✓ Star-based ranking and leaderboard system
- ✓ User advertising and promotion opportunities
- ✓ Profile customization and theme selection
- ✓ Ratings and reviews system
- ✓ Advertisement management tools
- ✓ User management and moderation features
- ✓ Responsive, interactive, and validation-driven user experience

Pejman MERN Wonderland is more than just a gaming platform—it is a community-driven ecosystem where entertainment, competition, advertising, and user engagement come together in one modern web application.

Use case Diagram





Special Features

Advanced Registration & Authentication Validation

The application includes a comprehensive registration system with multiple layers of validation:

- **Live Password Validation** – Passwords must contain at least:
 - 8 characters
 - 1 uppercase letter
 - 1 lowercase letter
 - 1 numeric digit
- **Live Password Confirmation** – Users must re-enter their password to ensure it was entered correctly.
- **Administrator Secret Key** – Users who wish to register as administrators must provide a special secret key, preventing unauthorized admin registrations.

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- **Unique Usernames** – Duplicate usernames are not allowed, ensuring that every account has a unique identity.

User Experience Enhancements

- **Registered User Counter** – The home page displays the current number of registered users.
- **Personalized Welcome Message** – Logged-in users are greeted with a customized welcome message.
- **Persistent Advertisement Display** – Advertisements remain visible throughout the application, including while users are playing games.
- **Profile Editing Validation** – The profile editing page includes the same live password validation and confirmation features available during registration.



Advertisement Management

- **Advertisement Creation & Editing Validation** – All advertisement forms include appropriate validation to ensure data integrity.
- **Image Upload Support** – Administrators must upload at least one image when creating an advertisement.
- **Cloud Storage Integration** – Advertisement images are securely stored using “Cloudinary”.
- **Automatic Image Cleanup** – When an advertisement is deleted, or images are removed during editing, the corresponding files are automatically removed from “Cloudinary” as well.

Administrative Tools

- **Live User Search** – Administrators can instantly search for users by username.
- **User Management** – Administrators can view, manage, and remove registered user accounts when necessary.



Intelligent Game Design

- **Strategic Opponents** – In games that involve playing against an opponent, the application implements intelligent strategies to create a more challenging and engaging experience.
- **Comprehensive Game Validation** – Every game includes dedicated validation logic, with unique error messages tailored to specific user mistakes.
- **Confirmation Dialogs** – Important actions such as resetting a game or returning to the home page require user confirmation to prevent accidental actions.
- **Detailed Game Instructions** – Every game includes a dedicated "About the Game" section explaining its rules, objectives, and difficulty levels.
- **Timer-Based Challenges** – Several games include countdown timers, requiring players to complete objectives within a limited time.
- **External API Integration** – Certain games use external APIs to dynamically generate game data and enhance replayability.

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- **Rich Visual Content** – The application includes a large variety of images, icons, and visual elements to create an engaging user experience.
 - **Advanced Algorithms & Data Structures** – Some games leverage sophisticated programming concepts and algorithms, including recursive solutions and other advanced data-structure techniques.

Community Features

- **Ratings & Reviews System** – Players can rate and review each game, helping other users discover the most enjoyable experiences while providing valuable feedback for future improvements.

Tools & Technologies Implemented

HTML / CSS / JavaScript / Bootstrap

Node.js / Express.js

React.js

MongoDB / MongoDB Compass

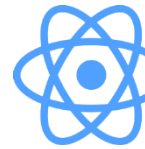
GitHub / GitHub Desktop / Git Bash

Postman

Visual Studio Code

Cloudinary

AI / ChatGPT



Future Enhancements

The project has been designed with scalability and extensibility in mind. The following features could be added in future versions to further enhance user engagement, community interaction, and platform functionality.

Enhanced Rating System

- Display the average rating for each game based on user reviews.
- Allow users to quickly identify the most popular and highly rated games.

Expanded Community Features

Introduce social interaction capabilities between users, including:

- Liking and disliking reviews.
- Replying to reviews and creating discussion threads.
- Sending stars to other users.
- Trading or gifting stars between players.
- Additional community-driven engagement features.



Detailed Player Statistics

- Display the number of stars earned for each game separately on the user's profile page.
- Provide detailed performance analytics and achievements.
- Allow users to identify their strongest and weakest games.

Personality Analysis (MBTI Integration)

- Add an MBTI personality assessment for users.
- Determine each user's personality type based on quiz results.
- Recommend games that best match the user's personality and strengths.
- Provide personalized gameplay insights and recommendations.



Advertiser Role & Business Collaboration System

Introduce a new user role: **Advertiser**.

This feature would include:

- A dedicated communication area for administrators and advertisers.
- Advertiser search capabilities for finding and contacting administrators.
- Advertisement proposal submissions.
- Commission negotiation between advertisers and administrators.
- Approval or rejection workflows for advertising requests.

Secure Payment & Transaction System

- Enable advertisers to transfer payments through the platform.
- Allow transactions only after an advertising proposal has been approved.
- Provide a secure and transparent payment process for both parties.



Game-Specific Leaderboards

- Display top-performing players for each game individually.
- Showcase rankings directly on each game's page.
- Encourage competition among players with game-specific achievements.

Life System

Introduce a life-based gameplay mechanism:

- Each player starts with a limited number of lives (e.g., 5 lives per day).
- Losing a game consumes one life.
- Players who run out of lives must wait until the next day before playing again.
- Encourage strategic gameplay and increase the value of victories.



Daily Lottery & Lucky Players Event

Implement a daily lottery system that randomly selects a group of lucky players.

For example:

- 10 users are randomly chosen each day.
- Stars earned by lucky users are automatically doubled for that day.
- Lucky users receive unlimited lives during the event period.
- Special rewards and bonuses can be introduced to increase engagement and daily participation.

Long-Term Vision

These enhancements would transform the platform from a gaming and advertising application into a comprehensive community-driven ecosystem, combining entertainment, competition, social interaction, business opportunities, and reward-based engagement.

Summary

Pejman MERN Wonderland is a full-stack MERN application that combines gaming, competition, and advertising into a single interactive platform. Users can play 21 games, earn stars, compete on leaderboards, customize profiles, publish advertising messages, and review games, while administrators manage advertisements, users, and platform content. The project showcases advanced full-stack development concepts including authentication, authorization, CRUD operations, Cloudinary integration, external APIs, game algorithms, form validation, and role-based access control.